Final Exam

- Due No due date
- Points 100
- Questions 7
- Available May 9 at 12am - May 14 at 5pm
- Time Limit 75 Minutes

Instructions

This exam is auto-graded as much as possible. All problems that are auto-marked incorrect will be manually examined by the teaching staff, and all auto-graded problems have space to show work. If an auto-graded problem is marked wrong and you provide correct work--that is, your error is demonstrably one of data entry--your points will be returned. You may forgo showing work; however, no points will be returned on problems where work is not provided.

All problems that specify memory quantities using SI prefixes are to be understood to use the base-2 versions of those prefixes. That is, k (kilo or kibi) represents a multiplication by 2^{10}, M (mega or mebi) by 2^{20}, etc. This will not be restated in the problems. SI prefixes on quantities that are not memory-related use base-10 values.

Bit ranges, when required, should be given inclusive on both ends; if you were asked to provide the bit ranges for IEEE754 32-bit floating point, sign would encompass bits 31-31, exponent 30-23, and mantissa 22-0.

Correctly entered auto-graded answers should **never include units!** Units are specified for you. Please enter only a number. Pay attention to the given unit so that you may enter the correct value.

In all problems related to ARM assembly, integer values are always to be assumed to be 8 bytes each. This will not be restated in the problems.

You are permitted to use a simple calcuator to aid you with arithmetic. It should be installed as a program on the computer in the testing center. Ask a proctor if you have trouble finding it.

This quiz was locked May 14 at 5pm.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	60 minutes	94.5 out of 100

ⓘ Correct answers are hidden.

Score for this quiz: 94.5 out of 100
Submitted May 13 at 8:34pm

This attempt took 60 minutes.

Partial

Question 1

20 / 25 pts

Complete the following table converting between decimal and 32-bit IEEE 754. Give decimal answers exactly in base 10 (all values in the table have exact representations in both base 10 and IEEE 754). Give IEEE 754 answers as strings of 32 zeros and ones, split up in the boxes into sign bit, exponent, and mantissa ("mantissa" is also called "fraction"). In the binary fields, please be sure that you put the appropriate number of bits, even leading zeros, and use no characters other than zeros and ones. For the mantissa only, you may enter the full 23 bits or the all bits left of and including the least significant 1 (dropping the trailing zeros). If the mantissa is zero, you must enter either one or twenty-three zeros. Any variation from this input specification will cause Canvas's automatic grader to mark your answer wrong.

The first row in the table is completed as an example

Decimal	IEEE 754 Sign	IEEE 754 Exponent	IEEE 754 Mantissa
6.1875	0	10000001	100011 or 10001100000000000000000
3.09375	0	10000000	100011
-2	1	10000000	000000
1376	0	10001001	01011

Answer 1:
10000000
Answer 2:
1
Answer 3:
10000000
Answer 4:
000000
Answer 5:
1376

Question 2
4.5 / 0 pts

For the possibility of partial credit, show your work for the previous problem.

Your Answer:

For the first one to find the mantissa we convert the value to binary, 3 becomes 11 and the decimal value becomes 00011 which then becomes 11.00011
We then normalize it (only have 1 before the decimal point) which then equals 1.100011 multiplied by 2^1. 100011 becomes mantissa
To find the exponent we add bias to 1 which then equals 128. Then converting 128 to binary we get 10000000 which becomes the exponent.
Since the value is positive the sign bit is 0

For -2, sign bit becomes 1. Following the above steps on this, 2 becomes 0010.0, normalizing this we get 1.0000 multiplied by 2^1. Mantissa becomes all zeroes while exponent becomes 1 + 127 which is 128 then converting it to binary we get exponent 10000000

We use the formula x = (-1)^sign bit into (1.Fraction) into 2^exponent
we convert exponent to binary then subtract 127 from it to get the exponent. Then to find the fraction we convert 01011 to binary and get the fraction bit back. then this becomes 1376

Partial

Question 3
12.5 / 25 pts

An unpipelined processor can be pipelined by breaking it down into seven parallel stages: instruction fetch is two stages (IF1 and IF2), instruction decode is also two stages (ID1 and ID2), then execute (EX), memory access (MEM), and writeback (WB). These stages will require, respectively, 160, 145, 150, 155, 165, 175, and 150 ps.

Fill in the blanks below. Do not include units in your answer; units are already given for you.

What is the clock cycle time of the unpipelined processor? 1100 ps

What is the clock frequency of the unpipelined processor (**please round to two decimal places**)?

909.10 MHz

What is the clock cycle time of the pipelined processor? 175 ps

What is the clock frequency of the pipelined processor (**please round to two decimal places**)?

5.70 GHz

Answer 1:
1100
Answer 2:
909.10
Answer 3:
175
Answer 4:
5.70

Question 4
12.5 / 0 pts

For the possibility of partial credit, show your work for the previous problem.

Your Answer:

Clock Cycle of unpipelined processor is the sum of all the stages. Which add up to 160 + 145 + 150 + 155 + 165 + 175 + 150 which is 1100 ps
Mhz is 10^6. Frequency is 1 / cycle time. So for unpipelined processor this value becomes 909.10 or to satisfy the two decimal places either 909.09 or 909.10
The longest wait time is the ps for a pipelined processor which in this case is MEM which is 175 ps
Again similar to mhz we do the same steps for finding Ghz for pipelined processor. Ghz is 10^9. It then becomes 5.70 rounding to two decimal places.

Partial

Question 5
20 / 25 pts

You just got a new laptop with 32 GB RAM. Your operating system uses 48-bit virtual addresses, has 4 kB pages, and uses a hierarchical page table for translation from virtual to physical addresses.

Please give exact answers without units. If the answer is a power of two, Canvas will correctly mark either of the multiplied out integer value or the exponential in the form *2^n* (where n is the integer exponent).

What is the maximum number of pages that may exist in the virtual address space? 2^36

How many physical pages exist on this system? 2^23

The virtual address is broken into 5 fields, with bits 47-39 indexing the root page table, bits 38-30 indexing the second level (direcly under the root), 29-21 indexing the third level, and bits 20-12 indexing the fourth and final level. Bits 11-0 index a byte within a page.

A program uses the following virtual pages (virtual page numbers given in binary and broken into groups of 9 bits for readability, 36 bits are given because the low-order 12 bits are for indexing a byte within a page):

```
000000001 000000000 000000000 000000000
000000001 000000000 000000001 000000000
000000001 000000000 000000010 000000000
000000001 000000000 000000010 000001000
000000010 001110100 111101010 110010010
000000010 001110100 111101011 110010010
```

How many pages exist in first (root) level of the page table? 1

How many pages exist in the second level of the page table? 2

How many **kilobytes** of storage are needed to hold the page table? 36

Answer 1:
2^36
Answer 2:
2^23
Answer 3:
1
Answer 4:
2
Answer 5:
36

Question 6
0 / 0 pts

For the possibility of partial credit, show your work for the previous problem.

Your Answer:

32gb = 32 x 2^30 which becomes 2^35 bytes
Page size is 4kb which becomes 2 ^12 bytes

Number of virtual address pages = 2 ^ virtual address bits / page size. Which becomes 2^48/2^12 which = 2^36

Number of physical pages = ram size / page size. Which becomes 2^35/2^12 = 2^23

The rest of the answers are pretty self explanatory idk how to explain it.

Question 7
25 / 25 pts

A very small cache has the following configuration:

- 2-way associative, using an LRU replacement policy
- 8 sets
- 4 bytes per cache block

yielding a total capacity of 64 bytes. The cache is initially all blocks invalid. When both blocks in a set are invalid, new data prefer way 0.

An array of characters (one byte each) is located at address 0, such that the index of each character in the array is exactly the address of that character. Indices in the array are accessed in the order given below.

Select the appropriate choices from the dropdowns to give the index of the first element in each block (the base address of the block) of the cache after all of the accesses are complete. In order to help you get started, we've done the first 8 accesses for you, and to help you verify your work, some of the answers are provided for you; if your answers don't match the provided answers, you made an error in your calculations.

Addresses are accessed through the cache in the following order:

```
83, 119,  73,  27,  79,  85,  20,  84,
25,  14,  89,  69, 126,  85,  32,  77,
18,  10,  70,  92,  74,  72,  17, 108,
68,  82,  53,  85,  89, 112,  98,  44
```

Working through the first 8 accesses:

- 83 misses. It is added to set 4 in way 0.
- 119 misses. It is added to set 5 in way 0.
- 73 misses. It is added to set 2 in way 0.
- 27 misses. It is added to set 6 in way 0.

- 79 misses. It is added to set 3 in way 0.
- 85 misses. It is added to set 5 in way 1.
- 20 misses. It is added to set 5 in way 0, knocking out the block beginning at address 116.
- 84 hits in set 5, way 1.

At this point, the cache has the following state:

- Way 0 has data in sets 2, 3, 4, 5, and 6.
- Way 1 has data in set 5.
- Way 0 in sets 0, 1, and 7 are invalid, as are all sets in way 1 aside from set 5.
- In set 5, way 0 is least recently used.

Complete the state of the cache after the remaining 24 accesses. In those pulldowns which contain more than one address, the first number corresponds with way 0, the second with way 1.

- Set 0: The blocks begin at addresses [Select]
- Set 1: Way 1 is invalid. The block in way 0 begins at address [Select]
- Set 2: The blocks begin at addresses 72 and 8
- Set 3: Way 0 begins at address 44. Way 1 begins at address 108
- Set 4: The blocks begin at addresses 80 and 112
- Set 5: The blocks begin at addresses 52 and 84
- Set 6: Way 1 begins at address 88. Way 0 begins at address 24
- Set 7: The blocks begin at addresses 124 and 92

Answer 1:

0 and 32

0 and 96

32 and 96

64 and 96

Answer 2:

4

36

68

100

Answer 3:

40 and 8

40 and 104

72 and 8

72 and 40

Answer 4:

12

76

108

112

Answer 5:

16 and 80

16 and 112

48 and 112

80 and 112

Answer 6:

20 and 52

52 and 84

52 and 116

84 and 116

Answer 7:

24

25

56

120

Answer 8:

92 and 28

92 and 60

124 and 60

124 and 92

Quiz Score: 94.5 out of 100